



Translated

From: e-maxx.ru

# Prefix function. Knuth–Morris–Pratt algorithm

## Prefix function definition

You are given a string  $s$  of length  $n$ . The **prefix function** for this string is defined as an array  $\pi$  of length  $n$ , where  $\pi[i]$  is the length of the longest proper prefix of the substring  $s[0 \dots i]$  which is also a suffix of this substring. A proper prefix of a string is a prefix that is not equal to the string itself. By definition,  $\pi[0] = 0$ .

Mathematically the definition of the prefix function can be written as follows:

$$\pi[i] = \max_{\{k = 0 \dots i\}} \{k : s[0 \dots k-1] = s[i-(k-1) \dots i]\}$$

For example, prefix function of string "abcbcd" is  $[0, 0, 0, 1, 2, 3, 0]$ , and prefix function of string "aabaab" is  $[0, 1, 0, 1, 2, 3]$ .

## Trivial Algorithm

An algorithm which follows the definition of prefix function exactly is the following:

```
vector<int> prefix_function(string s) {
    int n = (int)s.length();
    vector<int> pi(n);
    for (int i = 0; i < n; i++)
        for (int k = 0; k <= i; k++)
            if (s.substr(0, k) == s.substr(i-k+1, k))
                pi[i] = k;
    return pi;
}
```

It is easy to see that its complexity is  $O(n^3)$ , which has room for improvement.

## Efficient Algorithm

This algorithm was proposed by Knuth and Pratt and independently from them by Morris in 1977. It was used as the main function of a substring search algorithm.